Effective-dimension bounds in the central limit theorem for centered Euclidean balls

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Gaussian approximation for sums

Independent centered vectors $X_1, \dots, X_n \in \mathbb{R}^p$:

$$S_n = \sum_{i=1}^{n} X_i, \qquad \Sigma = \mathrm{Var}(S_n), \qquad G \sim N(0, \Sigma).$$

For a class of sets $\mathcal{A}$, consider

$$\Delta_n(\mathcal{A}) = \sup_{A \in \mathcal{A}} \Big| \mathbb{P}(S_n \in A) - \mathbb{P}(G \in A) \Big|.$$

Which features of the covariance determine the error?

- The sample size n and the moments of the increments.
- The geometry of $\mathcal{A}$ and the spectrum of Σ.

The geometry of the test sets

Class	Typical statistic	Main geometric feature
Convex sets	General multivariate inference	Affine invariance
Euclidean balls	Norms and squared errors	Euclidean geometry
Rectangles	Coordinate maxima	Coordinatewise geometry

Our class is $\mathcal{B}_0 = \{B(0, R) : R \geq 0\}$. Only the radius varies.

Convex sets: explicit dimension dependence

For nonsingular Σ, write

$$\beta_3 = \sum_{i=1}^{n} \mathbb{E} \left\| \Sigma^{-1/2} X_i \right\|^3 .$$

Bentkus (2003, 2005)

$$\Delta_n(\mathcal{C}_p) \le C p^{1/4} \beta_3 .$$

For $X_i = Y_i/\sqrt{n}$, with $\mathrm{Var}(Y_i) = I_p$ and $\mathbb{E}\,\|Y_i\|^3 \asymp p^{3/2}$, this gives

$$\Delta_n(\mathcal{C}_p) \lesssim \frac{p^{7/4}}{\sqrt{n}} .$$

Whitened moments retain the contribution of every direction.

Bentkus (1986, 2003, 2005). The 2003 i.i.d. bound has constant 400.

Stein's method and fourth-moment bounds

- Götze (1991) and Raič (2019): multivariate normal approximation for nonsmooth sets, with explicit constants.
- Chen–Shao (2007), Chen–Fang (2015), Shao–Zhang (2022): concentration methods and nonlinear statistics.

Fang–Koike (2024)

For i.i.d. isotropic Y_i with $\mathbb{E}\,\|Y_i\|^4 \le Cp^2$:

$$\text{convex sets: } \frac{p^{5/4}}{\sqrt{n}}, \qquad \text{all balls: } \frac{p}{\sqrt{n}},$$

up to logarithmic factors.

A separate centered-ball bound in Fang–Koike also permits $p = o(n)$.

Centered balls and quadratic forms: faster rates

Esseen (1945)	$n^{-p/(p+1)}$ for centered, axis-aligned ellipsoids under fourth moments
Bentkus–Götze (1996, 1997)	n^{-1} rates for quadratic forms under dimension conditions
Götze–Zaitsev (2014)	n^{-1} for nondegenerate quadratic forms in fixed $p \geq 5$, under fourth moments

The Götze–Zaitsev constants depend on dimension and covariance.

Our objective: uniform spectral control of the n^{-1} bound.

Why centering the ball matters

In a formal Edgeworth expansion, the first correction has the form

$$n^{-1/2} P_1(x)\varphi_\Sigma(x), \qquad P_1(-x) = -P_1(x).$$

Since a centered ball is symmetric,

$$\int_{\|x\| \le R} P_1(x)\varphi_\Sigma(x)\, dx = 0.$$

- The usual first correction vanishes.
- The remainder still needs uniform control, including lattice effects.

This is an explanation of the rate, not an assumption of an Edgeworth density expansion.

Hilbert-space balls and covariance eigenvalues

Zalesskii–Sazonov–Ulyanov (1991)

- Third-moment bounds of order $n^{-1/2}$ for balls.
- Constants involve the first six covariance eigenvalues and the center.

Nagaev–Chebotarev (2007)

- Study the second Edgeworth term and its spectral dependence.
- Their conjecture concerns dependence on leading eigenvalues in a short-expansion error bound.

The number of eigenvalues in a bound and the aggregate condition $v \geq 6$ are distinct.

Rectangles: logarithmic dimension dependence

Let $S_n = n^{-1/2} \sum_i Y_i$, with coordinate variances normalized on average and $|Y_{ij}| \le B_n$.

Chernozhukov–Chetverikov–Kato (2017)	$C \left(\dfrac{B_n^2 \log^7(pn)}{n} \right)^{1/6}$
Chernozhukov–Chetverikov–Kato–Koike (2022)	$C \left(\dfrac{B_n^2 \log^5(pn)}{n} \right)^{1/4}$
Chernozhukov–Chetverikov–Koike (2023)	$C \left(\dfrac{B_n^2 \log^3 p}{n} \right)^{1/2} \log n$

The first two allow singular covariance. The third display assumes a fixed positive lower bound on the correlation matrix's smallest eigenvalue.

Other fast results allow singular covariance under additional structure.

Effective dimension in statistics

- **Covariance estimation.** Koltchinskii–Lounici (2017) express operator-norm errors through effective rank.
- **Spectral projectors.** Effective rank appears alongside spectral gaps and variance parameters.
- **Regression.** Cheng–Montanari (2024) and Bach (2024) describe high-dimensional risk through covariance spectra and regularization.

These results motivate distributional bounds that reflect how much variance the coordinates collectively carry.

The effective dimensions used in different statistical problems need not coincide.

Two spectral quantities

For a nonzero covariance operator, define

$$\boxed{r(\Sigma) = \frac{\operatorname{tr}\Sigma}{\|\Sigma\|}}, \qquad \boxed{v(\Sigma) = \frac{\operatorname{tr}(\Sigma^2)}{\|\Sigma\|^2}}.$$

With $\|\Sigma\| = 1$ and eigenvalues $s_j \in [0, 1]$,

$$r = \sum_j s_j, \qquad v = \sum_j s_j^2, \qquad 1 \le v \le r \le p.$$

- Both quantities are invariant under scalar rescaling.
- For $\Sigma = I_p$, one has $r = v = p$.
- Many small eigenvalues may contribute little to either sum.

A spectrum with bounded effective rank

Take six eigenvalues equal to one and all remaining eigenvalues equal to $\varepsilon = 6/(p-6)$, for $p \geq 12$.

$$r = 12, \qquad v = 6 + \frac{36}{p-6}, \qquad r \leq 2v.$$

p	Smallest eigenvalue	r	v
66	0.1	12	6.6
606	0.01	12	6.06
6006	0.001	12	6.006

The spectral constants stay fixed as p grows and the tail eigenvalues shrink.

Balance and directional fourth moments

Normalize $\|\Sigma\| = 1$ and write $\Sigma_i = \mathrm{Var}(X_i)$.

(A1) No increment dominates the operator norm or trace

$$n \, \|\Sigma_i\| \le C_1, \qquad n \, \mathrm{tr}\, \Sigma_i \le C_1 r.$$

(A2) Uniform directional fourth moments

$$\mathbb{E}|u^\top X_i|^4 \le K_4 (u^\top \Sigma_i u)^2 \qquad \text{for every } u \in \mathbb{R}^p.$$

For i.i.d. normalized sums, (A1) holds with $C_1 = 1$. The current theorem requires no exponential moments.

Spectral assumptions: (A3)(i) and (A3)(ii)

(i) Trace and squared spectral mass are comparable

$$r \le C_\Sigma v, \qquad C_\Sigma \ge 1 \text{ fixed.}$$

The ratio is a variance-weighted average:

$$\frac{v}{r} = \sum_j \frac{s_j}{\sum_\ell s_\ell} s_j \ge C_\Sigma^{-1}.$$

(ii) The squared spectral mass satisfies $v \ge 6$

This condition is stronger than having six positive eigenvalues. It permits arbitrarily small additional eigenvalues.

Main theorem

Under (A1), (A2), and (A3),

$$\Delta_n(\mathcal{B}_0) = \sup_{z \geq 0} \left| \mathbb{P}(\|S_n\|^2 \leq z) - \mathbb{P}(\|G\|^2 \leq z) \right|$$

$$\boxed{\Delta_n(\mathcal{B}_0) \leq C \, \frac{r(\Sigma)^{3/2}}{n}.}$$

- C depends only on C_1, K_4, C_Σ.
- Uniform at $v = 6$, with no logarithmic loss.
- $r = o(n^{2/3})$ suffices for convergence, even when $p \gg n$.

Naumov–Samsonov–Spokoiny–Ulyanov, current manuscript, Theorem 2.1.

An isotropic comparison

For i.i.d. Y_i with $\mathrm{Var}(Y_i) = I_p$, a fixed directional fourth-moment bound, and $p \geq 6$, one has $r = v = p$.

Result	Test sets	Bound's order
Bentkus	Convex sets	$p^{7/4}/\sqrt{n}$
Bentkus	All balls	$p^{3/2}/\sqrt{n}$
Fang–Koike	All balls	$p/\sqrt{n}$ (logs suppressed)
Present theorem	Centered balls	$p^{3/2}/n$

The faster rate uses a smaller class of sets. For nonisotropic spectra, our dimension factor becomes $r^{3/2}$.

Why the comparison $r \leq C_\Sigma v$ matters

Fix $k \geq 6$ and take

$$\Sigma_n = \mathrm{diag}(I_k, n^{-3/2} I_{n^2}), \qquad r_n = k + \sqrt{n}, \quad v_n = k + \frac{1}{n}.$$

Let $X_{n,i} = \eta_i Z_{n,i}/\sqrt{n}$, where $Z_{n,i} \sim N(0, \Sigma_n)$ and $\eta_i \in \{0, \sqrt{2}\}$ with equal probabilities, independently.

$$\frac{r_n^{3/2}}{n} \longrightarrow 0, \qquad \liminf_{n\to\infty} \Delta_n(\mathcal{B}_0) > 0.$$

(A1), (A2), and $v_n \geq 6$ hold, but (A3)(i) fails.

Current manuscript, Proposition 2.2. The projections are uniformly sub-Gaussian.

The lower spectral threshold

In dimension four, take independent Rademacher coordinates:

$$Y_i = (\varepsilon_{i1}, \ldots, \varepsilon_{i4}), \qquad S_n = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} Y_i.$$

Then $\Sigma = I_4$, $r = v = 4$, and

$$\limsup_{n \to \infty} n \Delta_n(\mathcal{B}_0) = \infty.$$

- Lattice mass on individual spheres obstructs a uniform C/n bound.
- This example leaves the case $v = 5$ unsettled for our uniform theorem.

Current manuscript, Proposition 2.3. Fixed-dimensional $p = 5$ rates are known.

The same formulation in a Hilbert space

Let X_i take values in a separable real Hilbert space and let Σ be trace class. Under the operator versions of (A1)–(A3),

$$\Delta_n(\mathcal{B}_0) \le C \frac{r(\Sigma)^{3/2}}{n}.$$

An infinite-dimensional spectrum

$$s_1 = \cdots = s_6 = 1, \qquad s_{6+j} = 2^{-j} \quad (j \ge 1).$$

Here

$$r = 7, \qquad v = 6 + \frac{1}{3},$$

so both spectral quantities stay finite although the rank is infinite.

Statistical uses of the bound

Confidence balls for a mean

Let $q_{1-\alpha}$ be the $(1-\alpha)$ quantile of $\|G\|$. For i.i.d. observations with mean μ, under the assumptions,

$$\left|\mathbb{P}\Big(\sqrt{n}\left\|\bar{Y}-\mu\right\| \le q_{1-\alpha}\Big) - (1-\alpha)\right| \le C\frac{r^{3/2}}{n}.$$

- Norm statistics in functional and kernel models.
- Quadratic approximations to likelihood and spectral-estimation errors.

Estimated covariance, bootstrap calibration, and nonlinear remainders need separate bounds.

Proof architecture

For characteristic functions f, f_G of the half squared norms,

$$\Delta_n \le C \int_0^U \frac{|f(t) - f_G(t)|}{t}\, dt + \frac{C}{U\sqrt{v}}.$$

1. Truncate and center using directional fourth moments.
2. Compare at moderate frequencies, retaining covariance structure.
3. Control high frequencies by symmetrization and arithmetic estimates.

Esseen smoothing. Götze–Zaitsev (2014) supplies the Fourier framework. Götze–Naumov–Spokoiny–Ulyanov (2019) supplies Gaussian density and comparison estimates.

The endpoint $v = 6$

For $v \geq 7$

The Fourier comparison has a fixed integrability margin.

For $6 \leq v < 7$

Assumption (A3) gives

$$r \leq 7C_\Sigma, \qquad s_6 \geq \frac{1}{6C_\Sigma - 5}.$$

The proof retains an independent Gaussian component during replacement and uses a second-order comparison.

This yields the same n^{-1} rate uniformly at six.

Proof scheme: overview and notation

Normalize $\|\Sigma\| = 1$, and write

$$r = \operatorname{tr}\Sigma, \qquad v = \operatorname{tr}(\Sigma^2).$$

Goal:

$$\Delta_n(\mathcal{B}_0) = \sup_{z \geq 0} \left| \mathbb{P}(\|S_n\|^2 \leq z) - \mathbb{P}(\|G\|^2 \leq z) \right| \leq C \frac{r^{3/2}}{n}.$$

Common reduction

1. Esseen smoothing and frequency split.
2. Truncate and center the increments; compare covariances.
3. Reduce to a bounded normalized array.

Two spectral regimes

- $v \geq 7$: bounded-array Proposition 3.4, Appendix A, Appendix C.
- $6 \leq v < 7$: Appendix D, second-order replacement.

Section 3.1–3.3 + Appendices A–D of the manuscript.

Step 0: Esseen smoothing and cutoffs

Let $f, \tilde{f}$ be the characteristic functions of the half squared norms.

$$\Delta_n \le C \int_0^U \frac{|f(t) - \tilde{f}(t)|}{t}\, dt + \frac{C}{U\sqrt{v}}.$$

Choose

$$U = \frac{n}{r^2}, \qquad \tau = C_H \frac{n^{1/3}}{\sqrt{v}}, \qquad \ell = \min(U, \tau).$$

By (A3)(i),

$$\frac{C}{U\sqrt{v}} = C\frac{r^2}{n\sqrt{v}} \le C\frac{r^{3/2}}{n}.$$

For $v \ge 7$, split $[0, \ell]$, $[\ell, U]$, $[U, \infty)$. For $6 \le v < 7$, use $\tau = T_0 n^{1/3}$ and $U = n/\hat{r}^2$.

Section 3.2, equations (3.12)–(3.17); Section 3.3.

Step 1: truncation and centering

Set $E_i = \{\|X_i\| > \sqrt{r}\}$ and

$$\mu_i = \mathbb{E}[X_i \mathbf{1}_{E_i^c}], \quad Y_i = X_i \mathbf{1}_{E_i^c} - \mu_i, \quad T_i = \mathrm{Var}(Y_i), \quad \mu = \sum_i \mu_i.$$

For $\epsilon_n = 2K_4 C_1/n$ and large n,

$$\sum_i \mathbb{P}(E_i) \le \frac{K_4 C_1^2}{n}, \qquad \|Y_i\| \le 2\sqrt{r},$$

$$(1-\epsilon_n)\Sigma_i \preceq T_i \preceq \Sigma_i, \qquad \|\Sigma^{-1/2}\mu\| \le \frac{K_4 C_1^{3/2}}{n}.$$

Directional fourth moments are preserved:

$$\mathbb{E}|u^\top Y_i|^4 \le 16 K_4 (1-\epsilon_n)^{-2} (u^\top T_i u)^2.$$

Lemma 3.1. No exponential moments are required.

Step 2: covariance transfer and reduction

Let

$$\Sigma' = \sum_i T_i, \quad q = \|\Sigma'\|, \quad \hat{X}_i = Y_i/\sqrt{q}, \quad \hat{\Sigma} = \Sigma'/q.$$

Then

$$(1-\epsilon_n)r \le \hat{r} \le \frac{r}{1-\epsilon_n}, \qquad (1-\epsilon_n)^2 v \le \hat{v} \le \frac{v}{(1-\epsilon_n)^2},$$

$$\hat{v} \ge \frac{11}{2}, \qquad s_6(\hat{\Sigma}) \ge (1-\epsilon_n)s_6(\Sigma).$$

Transfer:

$$\Delta_n(\mathcal{B}_0) \le \hat{\Delta}_n(\mathcal{B}_0) + \frac{C}{n} + C\frac{r}{n\sqrt{v}}.$$

The last term is $\le Cr^{3/2}/n$ by $r \le C_\Sigma v$.

Lemma 3.3; density bound Lemma 3.2 gives $\|f_{\|G\|}\|_\infty \le C\sqrt{r/v}$.

Regime $v \geq 7$: bounded array

After truncation one has $\hat{v} \geq 13/2$ and $\|\hat{X}_i\| \leq L_0\sqrt{\hat{r}}$. **Proposition 3.4** For bounded normalized arrays with $v \geq 13/2$,

$$\Delta_n(\mathcal{B}_0) \leq C\frac{r^{3/2}}{n}.$$

Proof uses $U = n/r^2$, $\tau = C_H n^{1/3}/\sqrt{v}$, and

$$\int_0^\tau \frac{|f - \tilde{f}|}{t}\, dt \leq C\frac{r^{3/2}}{n} \quad \text{(Prop. A.16)},$$

$$\int_\tau^U |f(t)|\frac{dt}{t} \leq \frac{C}{n} \quad \text{(Prop. C.2)}.$$

Gaussian tail:

$$D(t) = |\mathbb{E}e^{it\|G\|^2/2}| \leq (1+t^2)^{-v/4}, \qquad \int_\tau^\infty D(t)\frac{dt}{t} \leq \frac{C}{n}.$$

Proposition 3.4 and equations (3.15)–(3.17).

Moderate frequencies: Appendix A

Let $h(t) = n/(t^2 + |t|)$. For $h(t) \geq C(r+1)$,

$$|f(t) - \tilde{f}(t)| \leq CH(t) \left\{ \frac{r^2}{n} t^2 + \frac{r^3}{n} t^3 \right\},$$

where

$$H(t) = (1 + c_* t^2)^{-\kappa v/4} + Ce^{-ch(t)} + C \frac{r^{3/2}|t|^3}{n^2}.$$

At $\tau = C_H n^{1/3}/\sqrt{v}$, integration gives

$$\int_0^\tau \frac{|f - \tilde{f}|}{t}\, dt \leq C \frac{r^{3/2}}{n}.$$

Three contributions

- determinant: $Cn^{-1}\{r^2 v^{-1} + r^3 v^{-3/2}\}$;
- exponential: negligible by $h(\tau) \gg r + \log n$;
- polynomial: controlled by (A3)(i).

Appendix A: partition and moment matching

Ordering and partition Order increments so prefix covariances approximate $k\Sigma/n$ (Lemma A.8). Split indices into $L = 14$ classes with

$$0 \le Q_\ell \le \Sigma, \qquad \|LQ_\ell - \Sigma\|_{\mathrm{HS}} \le \epsilon.$$

Pair classes into seven independent blocks. **Moment-matching proxy** Construct $Z_i = A_i + G_i$ with $\mathrm{Var}(A_i) = \mathrm{Var}(G_i) = \Sigma_i/2$ and matched moments through order three. Telescoping Taylor expansion through degree three gives

$$f(t) - f_Z(t) = \sum_i \mathbb{E}\{\mathcal{R}_4(F_i, X_i) - \mathcal{R}_4(F_i, Z_i)\}.$$

Retain an independent Gaussian component in every replacement background; derivative estimates through order six use only fourth moments. This yields the pointwise bound (A.36).

Lemmas A.6, A.8–A.15; Proposition A.16.

High frequencies: arithmetic/lattice estimate

After the cutoff, use symmetrization:

$$f(t) = \mathbb{E}_{\mathcal{D}} \Psi_{\mathcal{D}}(t),$$
$$|\Psi_{\mathcal{D}}(t)| \le 2^{-1/2} \{F_{12}(t)^{1/2} + F_{13}(t)^{1/2}\}.$$

The real symmetrization inequality reduces the problem to bilinear forms in Rademacher sums. Select random frames of size $s = 6$ with singular values in $[a_0\sqrt{v}, a_1\sqrt{v}]$ (Appendix B). Two Hölder applications with exponent $q \asymp M/4$ reduce to a fixed 6×6 matrix kernel:

$$K_M(u; A) = \mathbb{E} e^{iuR_M^\top A R'_M}.$$

Lemma C.1: for $\beta = 1/2$, $\beta s = 3 > 2$,

$$\int_{u_0}^1 K_M(u; A)^\beta \frac{du}{u} \le \frac{C}{M}, \qquad u_0 = M^{-1+1/(\beta s)}.$$

Consequently

$$\int_U^{} |f(t)| dt \le C$$

Endpoint $6 \le v < 7$: Appendix D

Here $r \le 7C_\Sigma$ and

$$s_6(\Sigma) \ge \frac{1}{6C_\Sigma - 5} \quad \text{(Lemma D.1)}.$$

After truncation, $\hat{v} \ge 11/2$ and the sixth eigenvalue remains bounded below. Choose $\tau = T_0 n^{1/3}$. Need

$$\int_0^\tau \frac{|f_Z - \tilde{f}|}{t}\, dt \le \frac{C}{n} \quad \text{(Prop. D.2, D.5)}.$$

Second-order replacement Use the exact identity

$$\prod_i \mu_i - \prod_i \nu_i = \sum_i \delta_i \prod_{k \ne i} \nu_k + \sum_{i<j} \delta_i \delta_j \prod_{k<j} \nu_k \prod_{k>j} \mu_k.$$

The linear term retains Gaussian covariance $(\Sigma - \Sigma_i)/2$. The quadratic term gains n^{-1} and is controlled by eighth derivatives from fourth moments. This gives the same n^{-1} rate uniformly at $v = 6$, without a logarithmic loss.

Appendix D, Section 3.3.

Lower bounds, Hilbert extension, summary

Necessity of assumptions

- Proposition 2.2: (A1), (A2), (A3)(ii) hold, but (A3)(i) fails; $r_n/v_n \to \infty$, $r_n^{3/2}/n \to 0$, yet $\liminf \Delta_n(\mathcal{B}_0) > 0$.
- Proposition 2.3: $v = 4$; Rademacher lattice mass gives $\limsup n\Delta_n(\mathcal{B}_0) = \infty$.

Hilbert space Finite-rank approximation plus an extra Gaussian coordinate of variance $t_N = \sqrt{(6 - v_N)_+}$ preserves (A1)–(A3), then pass to the limit.

Where the assumptions are used (A1) balance; (A2) directional fourth moments; (A3)(i) transfer and spectral summability; (A3)(ii) determinant integrability and the endpoint. The proof combines Esseen smoothing, truncation, covariance partitioning, moment matching, frame/lattice estimates, and a second-order endpoint expansion.

Theorem 2.1; Propositions 2.2–2.3; Appendices A–F.

Conclusions and questions

$$\Delta_n(\mathcal{B}_0) \le C\frac{r(\Sigma)^{3/2}}{n}, \qquad v(\Sigma) \ge 6.$$

- The bound retains Euclidean geometry and uses aggregate spectral quantities.
- Balanced increments and directional fourth moments suffice.
- The result includes $p \gg n$ and trace class Hilbert-space models.

Further questions: uniform bounds near $v = 5$, weaker spectral balance, and feasible statistical calibration.

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